

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 14

Section 1 : MCQ

1. What will be the output of the following program?

```
class Main {  
    public static void main(String[] args) {  
        String s = new String("5");  
        System.out.println(1 + 1111 + s + 1 + 1010);  
    }  
}
```

Answer

1112511010

Status : Correct

Marks : 1/1

2. Predict the output for the following code:

```
class Main {  
    public static void main(String args[]) {  
        StringBuffer sb = new StringBuffer("I Java!");  
        sb.insert(5, "like ");  
        System.out.println(sb);  
    }  
}
```

Answer

I Javlike a!

Status : Correct

Marks : 1/1

3. What will be the output of the following code?

```
class Main {  
    public static void main(String args[]) {  
        String s1 = "Hello i love java";  
        String s2 = new String(s1);  
        System.out.println((s1 == s2) + " " + s1.equals(s2));  
    }  
}
```

Answer

false true

Status : Correct

Marks : 1/1

4. What will be the output of the following program?

```
class Main {  
    public static void main(String[] args) {  
        String greet = "Welcome\n";  
        System.out.print("String: " + greet);  
        int length = greet.length();  
        System.out.print("Length: " + length);  
    }  
}
```

Answer

String: WelcomeLength: 7

Status : Wrong

Marks : 0/1

5. Predict the output for the following code.

```
public class Main {  
    public static void main(String[] args) {  
        String a = "java";  
        char temp = a.charAt(1);  
        System.out.println(temp);  
    }  
}
```

Answer

a

Status : Correct

Marks : 1/1

6. What will be the output of the following program?

```
class Main {  
    public static void main(String args[]) {  
        String name="Work Hard";  
        name.concat("Success");  
        System.out.println(name);  
    }  
}
```

Answer

Work Hard

Status : Correct

Marks : 1/1

7. What will be the output of the following program?

```
class Main {
```

```
public static void main(String[] args) {  
    String s1 = "EDUCATION";  
    String s2 = new String("EDUCATION");  
    String s3 = "EDUCATION";  
    if (s1 == s2) {  
        System.out.println("s1 and s2 equal");  
    }  
    else {  
        System.out.println("s1 and s2 not equal");  
    }  
    if (s1 == s3) {  
        System.out.println("s1 and s3 equal");  
    }  
    else {  
        System.out.println("s1 and s3 not equal");  
    }  
}
```

Answer

s1 and s2 not equals1 and s3 equal

Status : Correct

Marks : 1/1

8. Predict the output for the following code:

```
public class Main {  
    public static void main(String[] args) {  
        float a = 10.0f;  
        String temp = Float.toString(a);  
        System.out.println(temp);  
    }  
}
```

Answer

10.0

Status : Correct

Marks : 1/1

9. What is the output of the following code?

```
class Main
{
    public static void main(String args[])
    {
        StringBuffer c = new StringBuffer("Hello");
        c.delete(0,2);
        System.out.println(c);
    }
}
```

Answer

llo

Status : Correct

Marks : 1/1

10. What will be the output for the following code?

```
class Main {
    public static void main(String[] args) {
        String languages[] = { "C", "C++", "Java", "Python", "Ruby"};
        for (String sample: languages) {
            System.out.println(sample);
        }
    }
}
```

Answer

CC++JavaPythonRuby

Status : Correct

Marks : 1/1

11. Predict the output for the following code.

```
class Main {
    public static void main(String[] fruits) {
        String fruit1 = new String("apple");
        String fruit2 = new String("orange");
    }
}
```

```
String fruit3 = new String("pear");
fruit3 = fruit1;
fruit2 = fruit3;
fruit1 = fruit2;
System.out.println(fruit1);
System.out.println(fruit2);
System.out.println(fruit3);
    }
}
```

Answer

appleappleapple

Status : Correct

Marks : 1/1

12. What will be the output of the following code?

```
class Main {
    public static void main(String args[]) {
        char c[] = {'j', 'a', 'v', 'a'};
        String s1 = new String(c);
        String s2 = new String(s1);
        System.out.println(s1);
        System.out.println(s2);
    }
}
```

Answer

javajava

Status : Correct

Marks : 1/1

13. What will be the output of the following code?

```
class Main {
    public static void main(String args[])
    {
        StringBuffer sb = new StringBuffer("Hello");
        System.out.println("buffer before = " + sb);
    }
}
```

```

        System.out.println("charAt(1) before = " + sb.charAt(1));
        sb.setCharAt(1, 'i');
        sb.setLength(2);
        System.out.println("buffer after = " + sb);
        System.out.println("charAt(1) after = " + sb.charAt(1));
    }
}

```

Answer

buffer before = Hello charAt(1) before = e buffer after = Hi charAt(1) after = i

Status : Correct

Marks : 1/1

14. What will be the output of the following program?

```

class Main {
    public static void main(String args[]) {
        StringBuffer sb = new StringBuffer("Hello");
        System.out.println("buffer = " + sb);
        System.out.println("length = " + sb.length());
        System.out.println("capacity = " + sb.capacity());
    }
}

```

Answer

buffer = Hello length = 5 capacity = 21

Status : Correct

Marks : 1/1

15. What will be the output of the following program?

```

public class Main {
    public static void main(String[] args) {
        String str = "1234.34";
        int a = Integer.parseInt(str);
        System.out.println(a);
    }
}

```

Answer

NumberFormatException

Status : Correct

Marks : 1/1

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 4_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

In a publishing company, editors often need to quickly analyze passages of text to check for punctuation usage. To assist them, you are asked to write a program that counts the number of specific punctuation marks in each passage.

The punctuation marks of interest are:

Commas (,) Periods (.) Question marks (?)

Input Format

The first line of input contains an integer T, representing the number of test cases (passages).

Each of the next T lines contains a single passage of text.

Output Format

For each test case, print three integers separated by spaces, representing the number of commas, periods, and question marks in the passage.

The first line of output corresponds to the first passage, the second line to the second passage, and so on.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

Hello, world. How are you?

Output: 1 1 1

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int t=sc.nextInt();
        sc.nextLine();
        for(int i=0;i<t;i++){
            String pass=sc.nextLine();
            int c=0,f=0,q=0;
            for(char ch: pass.toCharArray()){
                if(ch==','){
                    c++;
                }
                else if(ch=='.'){
                    f++;
                }
                else if(ch=='?'){
                    q++;
                }
            }
            System.out.println(c+" "+f+" "+q);
        }
    }
}
```

}
}
Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 4_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Anu is developing a tool for a conference registration system. Participants submit keywords related to their fields of interest. The organizer wants to sort these keywords alphabetically to generate tags for session grouping.

Write a program that accepts at least five keywords as input arguments and outputs them in sorted alphabetical order.

Input Format

The first line of input contains an integer n, representing the number of keywords.

The second line of input contains n space-separated keywords (string).

Output Format

The output prints n space separated strings representing the sorted keyword in alphabetical order.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

Blockchain Cloud AI Data Cybersecurity

Output: AI Blockchain Cloud Cybersecurity Data

Answer

```
import java.util.Scanner;
import java.util.Arrays;
class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
        String[] keywords=sc.nextLine().split(" ");
        Arrays.sort(keywords);
        for(String key:keywords){
            System.out.print(key+" ");
        }
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 4_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Bechan Chacha is seeking help to filter out valid mobile numbers from a list provided by his crush. He can only pick his crush's number if the list contains valid mobile numbers.

A mobile number is considered valid if:

It has exactly 10 digits. It consists only of numeric values (0–9). It does not begin with zero.

Your task is to determine whether each mobile number in the list is valid or not.

Input Format

The first line contains an integer T, representing the number of mobile numbers

to check.

The next T lines each contain a string S, representing a mobile number.

Output Format

For each mobile number S, the output print "YES" if it is valid.

Otherwise, print "NO".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1
9876543210

Output: YES

Answer

```
// You are using Java
import java.util.*;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
        for(int i=0;i<n;i++){
            String number=sc.nextLine();
            if(number.length()==10&&number.matches("[1-9][0-9]{9}")){
                System.out.println("YES");
            }
            else{
                System.out.println("NO");
            }
        }
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 4_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Arjun is learning how to filter words from a sentence based on grammar rules. He wants to identify the valid words in a sentence.

A word is considered valid if it satisfies all these conditions:

The word contains only alphabets (a-z, A-Z). The word length is at least 2 characters. The word should not contain digits or special characters.

Your task is to read a sentence and print all the valid words in it.

Input Format

The input contains a single line containing a sentence S.

Output Format

The output prints all the valid words separated by spaces.

If no valid word exists, print "No valid words."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Hello world1 123 ab" @\$ Hi

Output: Hello Hi

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        String[] words=sen.split(" ");
        boolean validword=false;
        for(String w:words){
            if(w.length()>=2&&w.matches("[a-zA-Z]+")){
                System.out.print(w+" ");
                validword=true;
            }
        }
        if(!validword){
            System.out.print("No valid words.");
        }
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 4_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

In a secure banking system, customers are required to create PIN codes for accessing their accounts. The bank wants to validate these PIN codes before accepting them.

A PIN code is considered valid if:

It consists of exactly 4 digits. All characters must be numeric (0–9). It cannot contain all identical digits (e.g., 1111 is invalid).

Your task is to determine whether each PIN code in the list is valid or not.

Input Format

The first line of input contains an integer T, representing the number of PIN codes to check.

The next T lines each contain a string S, representing a PIN code.

Output Format

For each PIN code S, the output print "YES" if it is valid.

Otherwise, the output print "NO".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1234

Output: YES

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int t=sc.nextInt();
```

```
        sc.nextLine();
```

```
        for(int i=0;i<t;i++){
```

```
            String pin=sc.nextLine();
```

```
            if(pin.length()==4&&pin.matches("[0-9]+")&&!pin.matches("(\\d)\\1{3}")){
```

```
                System.out.println("YES");
```

```
            }
```

```
            else{
```

```
                System.out.println("NO");
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_PAH

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Riya is preparing a puzzle game for her friends. She wants to include a feature that highlights special words in a sentence — specifically, palindromic words (words that read the same forward and backward).

Your task is to help Riya by writing a program that extracts all palindrome words from the given sentence. If there are no palindromes, print "No palindromes found".

Input Format

The input contains a single string S representing a sentence.

Output Format

The output prints all palindromic words separated by a space.

If no palindrome exists, print "No palindromes found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: madam went to school

Output: madam

Answer

```
// You are using Jav
import java.util.Scanner;
class main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        String[] words=sen.split(" ");
        boolean found=false;
        for(String word:words){
            String rev="";
            for(int i=word.length()-1;i>=0;i--){
                rev+=word.charAt(i);
            }
            if(word.equals(rev)){
                System.out.print(word+" ");
                found=true;
            }
        }
        if(!found){
            System.out.println("No palindromes found");
        }
    }
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

At a digital library, the system needs to analyze passages to identify the frequency of vowels, since they are key for linguistic research. You are asked to write a program that counts the number of vowels in each passage of text.

The vowels of interest are:

a, e, i, o, u (both uppercase and lowercase).

Input Format

The first line of input contains an integer T, representing the number of test cases (passages).

Each of the next T lines contains a single passage of text.

Output Format

For each test case, print a single integer representing the total number of vowels in the passage.

The first line of output corresponds to the first passage, the second line to the second passage, and so on.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1
Hello World

Output: 3

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int t=sc.nextInt();
        sc.nextLine();
```

```

for(int s=0;s<t;s++){
    String line=sc.nextLine();
    int count=0;
    for(int i=0;i<line.length();i++){
        char ch=line.charAt(i);
        if("aeiouAEIOU".indexOf(ch)!=-1){
            count++;
        }
    }
    System.out.println(count);
}
}
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Sana is analyzing text for a secret code. She wants to find all words in a sentence that start and end with the same letter. These words are considered "special words" for her analysis.

Your task is to write a program that extracts and prints all words that start and end with the same letter (case-insensitive).

If no such word exists, print "No special words found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words that start and end with the same letter separated by a space.

If no word satisfies the condition, print "No special words found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Anna went to the civic center

Output: Anna civic

Answer

```
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        String[] words=sen.split(" ");
        boolean f=false;
        for(String word:words){
            word=word.toLowerCase();
            if(word.length()>0&&word.charAt(0)==word.charAt(word.length()-1)){
                System.out.print(word+" ");
                f=true;
            }
        }
        if(!f){
            System.out.println("No special words found");
        }
    }
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Ravi is analyzing text messages for his research on typing patterns. He wants to count the number of uppercase letters, lowercase letters, and digits in a sentence to understand typing trends.

Your task is to help Ravi by writing a program that takes a sentence and prints the count of uppercase letters, lowercase letters, and digits.

Input Format

The input contains a single line containing a sentence (string).

Output Format

The output prints three integers separated by spaces:

- Number of uppercase letters
- Number of lowercase letters
- Number of digits

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Hello World 123

Output: 2 8 3

Answer

```
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        int uppercount=0;
        int lowercount=0;
        int digitcount=0;
        for(char ch: sen.toCharArray()){
            if(Character.isUpperCase(ch)){
                uppercount++;
            }
            else if(Character.isLowerCase(ch)){
                lowercount++;
            }
            else if(Character.isDigit(ch)){
                digitcount++;
            }
        }
        System.out.println(uppercount+" "+lowercount+" "+digitcount);
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Neha is analyzing text messages to identify words that have repeated characters. A word is considered "repetitive" if any character appears more than once in that word.

Your task is to write a program that extracts all words that contain repeated characters from a given sentence.

If no such word exists, print "No repetitive words found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words that contain repeated characters separated by a space.

If no word contains repeated characters, print "No repetitive words found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: letter balloon apple tree

Output: letter balloon apple tree

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        sc.close();
        String[] words=sen.split(" ");
        String res="";
        for(String word:words){
            boolean[] seen=new boolean[256];
            boolean repeated=false;
            for(char c:word.toCharArray()){
                if(seen[c]){
                    repeated=true;
                    break;
                }
                seen[c]=true;
            }
            if(repeated){
                res+=word+" ";
            }
        }
        if(res.length()>0){
            System.out.println(res.trim());
        }
        else{
```

```
System.out.println("No repetitive words found");
```

Status : Correct

Marks : 10/10

2. Problem Statement

In a college, students are required to create unique usernames for accessing the digital library.

The librarian needs your help to verify whether the usernames entered by students are valid.

A username is considered valid if:

It contains only letters (a–z, A–Z) and digits (0–9). Its length is between 5 and 15 characters (inclusive). It must start with a letter (not a digit).

Your task is to determine whether each username in the list is valid or not.

Input Format

The first line of input contains an integer T, representing the number of usernames to check.

The next T lines each contain a string S, representing a username.

Output Format

For each username S, the output print "YES" if it is valid.

Otherwise, the output print "NO".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

Alice123

Output: YES

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int t=sc.nextInt();
        sc.nextLine();
        for(int i=0;i<t;i++){
            String s=sc.nextLine();
            if(isValid(s)){
                System.out.println("YES");
            }
            else{
                System.out.println("NO");
            }
        }
    }
    static boolean isValid(String s){
        if(s.length()<5||s.length()>15)
            return false;
        if(!Character.isLetter(s.charAt(0)))
            return false;
        for(int i=0;i<s.length();i++){
            char ch=s.charAt(i);
            if(!Character.isLetterOrDigit(ch))
                return false;
        }
        return true;
    }
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Riya is preparing for a vocabulary test. Her teacher told her to focus on long words in her practice sentences, specifically words that have at least

5 letters.

Riya wants to write a program that will help her identify such words quickly.

Your task is to help Riya by printing all the words in a given sentence that have a length greater than or equal to 5.

If no such word exists, display "No long words found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words having length ≥ 5 , separated by a space.

If no such word is found, print "No long words found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The quick brown fox jumps over the lazy dog

Output: quick brown jumps

Answer

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        String[] words=sen.split(" ");
        boolean found=false;
        for(String word:words){
            if(word.length()>=5){
                System.out.print(word+" ");
                found=true;
            }
        }
    }
}
```

```
        if(!found){
            System.out.println("No long words found");
        }
    }
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Anjali is preparing a report on text complexity. She wants to identify all words in a sentence that contain at least one digit so she can analyze numeric mentions.

Your task is to write a program that extracts and prints all words containing at least one digit from a given sentence.

If no such word exists, print "No words with digits found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words containing at least one digit separated by a space.

If no word contains a digit, print "No words with digits found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The model X100 and Y200 are available

Output: X100 Y200

Answer

```
// You are using Java
import java.util.Scanner;
```

```
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        String sen=sc.nextLine();
        String[] words=sen.split(" ");
        boolean f=false;
        for(String word:words){
            for(char ch:word.toCharArray()){
                if(Character.isDigit(ch)){
                    System.out.print(word+" ");
                    f=true;
                    break;
                }
            }
        }
        if(!f){
            System.out.println("No words with digits found");
        }
    }
}
```

Status : Correct

Marks : 10/10